

A Rational-Octagon Counterexample to Necessity of the Weak Lindenstrauss–Perles Vertex Count

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Abstract

Mal, Paul, and Dey proved a vertex-count condition sufficient for a pair of finite-dimensional polygonal Banach spaces to have weak Lindenstrauss–Perles property. They proved necessity for a particular two-dimensional range and left necessity for arbitrary polygonal planes open. We give a negative answer. Let X be the polygonal plane whose unit ball is the rational octagon with vertices

$$\pm(1, 0), \quad \pm(2/3, 2/3), \quad \pm(0, 1), \quad \pm(-2/3, 2/3).$$

Then every extreme contraction $T : X \rightarrow X$ maps some vertex of B_X to a vertex of B_X , although B_X has eight vertices and fails the proposed strict count. The proof is a complete exact classification of the four-dimensional rational operator polytope: its 96 vertices form six signed-permutation orbits, each with an explicit vertex-to-vertex witness.

1 Source question and answer

A pair (X, Y) has *weak L-P property* if every extreme point T of the operator unit ball $B_{\mathcal{L}(X, Y)}$ satisfies

$$T(\text{Ext } B_X) \cap \text{Ext } B_Y \neq \emptyset.$$

The source’s Theorem 2.5 says that if $\dim X = n$, $\dim Y = m$, and $|\text{Ext } B_X| = 2(n + p)$, then

$$mp < n + p \tag{1}$$

is sufficient for weak L-P property. After proving necessity in one special two-dimensional range and constructing pair-dependent failures, the authors state that necessity for arbitrary polygonal planes is not known.

If $x \in M_I \cap \text{Ext}(B_{\mathbb{X}})$, then x is smooth point of $B_{\mathbb{Y}}$. Therefore, $I(\text{Ext}(B_{\mathbb{X}})) \cap \text{Ext}(B_{\mathbb{Y}}) = \emptyset$. Thus, the pair $(\mathbb{X}, \mathbb{Y})$ does not satisfy weak L-P property. This completes the proof of the theorem. $\square$

Although the above theorem indicates that the condition stated in Theorem 2.5 may be necessary for arbitrary two-dimensional polygonal Banach spaces $\mathbb{X}, \mathbb{Y}$ to satisfy weak L-P property, the answer is still not known in its full generality. However, if $\dim(\mathbb{X}) > 2$, then the condition is not necessary. We exhibit polyhedral Banach spaces $\mathbb{X}, \mathbb{Y}$ that satisfy weak L-P property but $\dim(\mathbb{X}) = n, \dim(\mathbb{Y}) = m, |\text{Ext}(B_{\mathbb{X}})| = 2(n + p)$ and $mp \geq n + p$ hold. To do so we need the following two lemmas.

Theorem 1 (Negative answer to necessity). *Let*

$$V_+ = \left\{ (1, 0), \left(\frac{2}{3}, \frac{2}{3} \right), (0, 1), \left(-\frac{2}{3}, \frac{2}{3} \right) \right\}, \quad B = \text{conv}(V_+ \cup -V_+),$$

and let $X = (\mathbb{R}^2, \|\cdot\|_B)$. Then (X, X) has weak L-P property. On the other hand, $|\text{Ext } B_X| = 8$, and the strict condition (1) fails.

Indeed, here $n = m = 2$ and $8 = 2(n + p)$ gives $p = 2$. Hence $mp = n + p = 4$. The rest of the packet proves the weak L-P assertion.

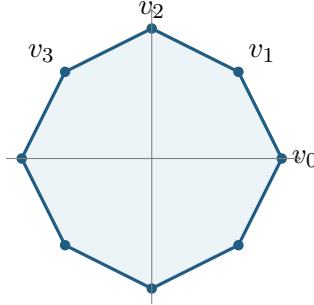
2 Proof intuition and the rational operator polytope

The universal quantifier over extreme contractions becomes finite here: $B_{\mathcal{L}(X, X)}$ is a four-dimensional rational polytope. We enumerate its vertices exactly, then use the octagon's signed-permutation symmetries to compress 96 cases to six matrices. A single vertex-to-vertex image for each representative propagates to its entire orbit.

The octagon and four representatives of its antipodal vertex pairs are shown below. Its eight facet functionals are

$$\mathcal{F} = \pm \left\{ (1, \tfrac{1}{2}), (\tfrac{1}{2}, 1), (-\tfrac{1}{2}, 1), (-1, \tfrac{1}{2}) \right\}.$$

Thus $B = \{x : f(x) \leq 1 \text{ for every } f \in \mathcal{F}\}$.



Identify an operator $A = (a_{ij})$ with $z = (a_{11}, a_{12}, a_{21}, a_{22}) \in \mathbb{R}^4$. Since B is the convex hull of $V_+ \cup -V_+$, the operator unit ball is exactly

$$K = B_{\mathcal{L}(X, X)} = \{A : f(Av) \leq 1 \text{ for every } v \in V_+, f \in \mathcal{F}\}. \quad (2)$$

This is a full-dimensional rational polytope cut out by $4 \cdot 8 = 32$ half-spaces.

Let G be the eight signed permutation matrices. Each element of G is an isometry of X , and $G \times G$ acts on K by $(U, W) \cdot A = UAW$.

Lemma 1 (Exact vertex certificate). *The polytope K has exactly 96 vertices. Under the $G \times G$ action they form six orbits, represented in Table 1.*

Why the enumeration is exhaustive

For $v = (v_1, v_2)$ and $f = (f_1, f_2)$, the inequality in (2) has normal

$$(f_1 v_1, f_1 v_2, f_2 v_1, f_2 v_2) \in \mathbb{Q}^4. \quad (3)$$

Every vertex of a full-dimensional polytope in $\mathbb{R}^4$ has four linearly independent active inequalities. We therefore examine all $\binom{32}{4} = 35,960$ four-subsets of the normals (3). For each subset, exact Gauss–Jordan

elimination over $\mathbb{Q}$ either detects dependence or returns its unique intersection. Retaining precisely the intersections that satisfy all 32 inequalities gives 96 distinct points. Conversely, every retained point is the unique feasible intersection of four independent supporting hyperplanes, hence is a vertex. Exact signed-permutation orbit closure yields sizes

$$8, 8, 16, 16, 16, 32,$$

and the six rational representatives in Table 1. This proves Lemma 1. The included verifier implements exactly these finite steps with Python’s standard-library `Fraction` class; no floating-point decision occurs.

Representative A	Orbit size	Vertex witness
$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	8	$Av_0 = -v_0$
$\begin{pmatrix} -2/3 & -2/3 \\ -2/3 & 2/3 \end{pmatrix}$	8	$Av_0 = -v_1$
$\begin{pmatrix} -1 & -1/2 \\ 0 & 0 \end{pmatrix}$	16	$Av_0 = -v_0$
$\begin{pmatrix} -2/3 & -1/3 \\ -2/3 & -1/3 \end{pmatrix}$	16	$Av_0 = -v_1$
$\begin{pmatrix} -3/4 & -3/4 \\ -1/2 & 1/2 \end{pmatrix}$	16	$Av_1 = -v_0$
$\begin{pmatrix} -3/4 & -2/3 \\ -1/2 & 2/3 \end{pmatrix}$	32	$Av_2 = v_3$
Total	96	

Table 1: The exact $G \times G$ orbit certificate.

3 Proof of the counterexample

Proof of Theorem 1. Extreme contractions $X \rightarrow X$ are precisely the vertices of K . By Lemma 1, every such vertex belongs to the orbit of one of the six matrices in Table 1. Each representative explicitly maps a vertex v_i to a vertex $\pm v_j$.

This property is invariant under the orbit action. Indeed, if $Av = w$ with $v, w \in \text{Ext } B$ and $U, W \in G$, then

$$(UAW)(W^{-1}v) = Uw,$$

and both $W^{-1}v$ and Uw are vertices of B . Therefore every extreme contraction maps some extreme point of B_X to an extreme point of B_X . Thus (X, X) has weak L-P property. The failure of (1) was computed after the statement of the theorem, completing the negative answer. $\square$

4 Reproduction and novelty check

From this packet directory, run

```
conda run --no-capture-output -n sandbox python \
  code/verify_rational_octagon_weak_lp.py
```

It reports 32 constraints, 35,960 four-subsets, 96 operator vertices, zero bad vertices, and the six symmetry orbits above.

A bounded 19 July 2026 search found the weak L-P terminology only in the 2019 introduction (arXiv:1902.06917), the source (arXiv:2006.15318), and a commented definition in arXiv:2407.05545 within the local 35,297-paper corpus. Exact web searches found no later answer or octagon example. Novelty confidence is moderate–high, subject to full bibliographic review.